

Bhargav Krishna Yadav Challa

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Education

National Institute of Technology, Nagpur, India

Jul 2020 – May 2024

Bachelor of Technology, Electrical and Electronics Engineering

Optional Subjects: Principles of Economics, Psychology, Human Resource Management

Work Experience

Quantum Research Intern - QuantumIntegrated Machines

March 2026 – Present

Public Code: github.com/bhargav2603/FMO-VQE | github.com/bhargav2603/Tomography

- Developed modular quantum-chemistry and measurement packages using **PySCF**, **Qiskit**, and **Qiskit Nature**, covering active-space Hamiltonians, UCCSD-VQE, FMO2 energy assembly, classical shadows, automated validation, testing, and resource analysis.
- Built an **FMO-VQE workflow** for $H_2/H_4/H_6$; FMO2 corrections reduced maximum absolute FCI deviation from **8.3 mHa** to **1.03 mHa**, bringing all corrected benchmark cases within the ± 1.6 mHa chemical-accuracy threshold.
- Validated UCCSD-VQE across **25 molecular simulator cases**, obtaining a maximum VQE-to-CASCI deviation of **0.382 mHa**; studied bond dissociation, optimizer convergence, active-space convergence, and solver-versus-truncation error.
- Implemented uniform and Hamiltonian-adaptive classical shadows, observing approximately $N^{-0.55}$ **error scaling** and up to **6.3 $\times$ lower sample variance** in a low-shot H_6 regime.

Associate – Cybersecurity Risk Consulting, PwC India

Oct 2024 – January 2026

- Conducted high-volume data analysis within a 24x7 SOC, analyzing **60+ security alerts daily** to identify critical cyber threats, and contributing to a **30% improvement** in incident response times.
- Utilized SIEM/SOAR tools (Chronicle, ArcSight) to streamline alert triage processes, enhancing response efficiency and ensuring **99% SLA adherence for high-severity threats**.
- Performed in-depth log analysis to identify patterns and correlations, producing clear, client-facing summaries with actionable recommendations that contributed to 30% faster threat resolution.
- Delivered **daily operational and log stoppage reports** to client teams, flagging non-reporting devices across 20+ critical assets to ensure 100% threat monitoring continuity.
- Evaluated client alert closures to validate response quality and recommend follow-ups, reducing repeat incidents by **5% month-over-month**.

Leadership Experience

Draftsperson, Student Governance Reform Initiative, NIT Nagpur

Jan 2024 – May 2024

- Drafted and submitted a **20+ page Student Constitution and Election Commission framework** to 3 college deans, setting the foundation for governance reform.
- Collected and translated qualitative insights from interviews into structured recommendations presented to the college deans impacting **3,500+ students**.
- Proposed 3+ new council roles to improve structure, accountability, and decision-making.

Co-Founder, VNIT_TSS Committee, NIT Nagpur

Aug 2023 – May 2024

- Founded a cultural body representing **800+ Telugu students**, enabling inclusive participation in campus life.
- Led planning and execution of **3 large-scale festivals (Sankranti & Ugadi '23-'24)** with a self-funded budget of **3 Lakh rupees**, both events had a footfall of **1200+ students**.

Student Mentor, NIT Nagpur

Aug 2022 – May 2024

- Mentored **15 first-year students** under the Student Mentorship Program (SMP), supporting academic performance, emotional wellbeing, and campus integration over 2 years.
- Handled high-impact mentorship cases, including mental health crisis support and **suicide prevention**, coordinating with faculty and counselors to ensure timely intervention and safety.
- Improved mentee retention and engagement by **30%** through regular check-ins and tailored guidance.

Founder, CTD (Code, Think, Develop) Tech Community, NIT Nagpur

Aug 2022 – May 2023

- Initiated an on-campus tech community to foster collaboration, mentorship, and project-based learning among like-minded peers. Initiative ran for **10 months but did not scale due to low engagement**.

Analytical & Professional Skills

- Cybersecurity:** SIEM, SOAR, Threat Detection, MITRE ATT&CK
- Quantum Computing:** Qiskit (Core, Nature, Aer)
- Programming:** Python
- Professional Certifications:** Communication Strategy (PwC), Network Security (LinkedIn Learning)

Additional Information

- Volunteered** with Pryaas, the college's social service club.
- Recently **developed a strong interest in reading**, exploring philosophy, nonfiction, and fiction.
- Passionate about world cinema and enjoy **exploring storytelling styles** across different cultures and genres.
- Have a deep appreciation for **Indian classical music and dance**; for their rhythm, history, and expressive depth.